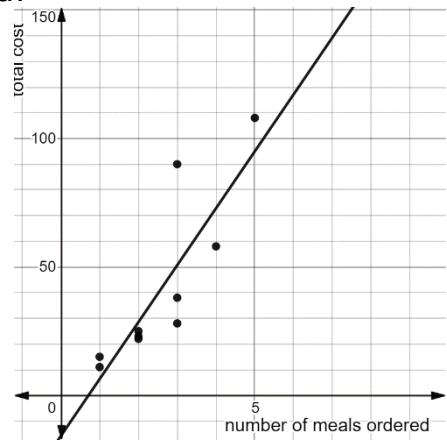


A restaurant took a random sample of customers ordering takeout meals. The data from the sample is shown.

Number of Meals ordered	1	2	5	3	2
Total Cost	\$15	\$23	\$108	\$28	\$22
Number of Meals ordered	4	3	3	1	2
Total Cost	\$58	\$90	\$38	\$11	\$25

This data is represented by a scatter plot with the function $c(x) = 22.17x - 15.83$ fit to the data, where c represents the total cost of the order and x represents the number of meals ordered.



For questions 2 through 6, complete the tables to find the residual values for each number of meals ordered.

1.

Number of Meals ordered	Actual Cost	Predicted Cost	Residual Values
1	\$15		
1	\$11		

2.

Number of Meals ordered	Actual Cost	Predicted Cost	Residual Values
2	\$23		
2	\$22		
2	\$25		

3.

Number of Meals ordered	Actual Cost	Predicted Cost	Residual Values
3	\$28		
3	\$90		
3	\$38		

4.

Number of Meals ordered	Actual Cost	Predicted Cost	Residual Values
4	\$58		

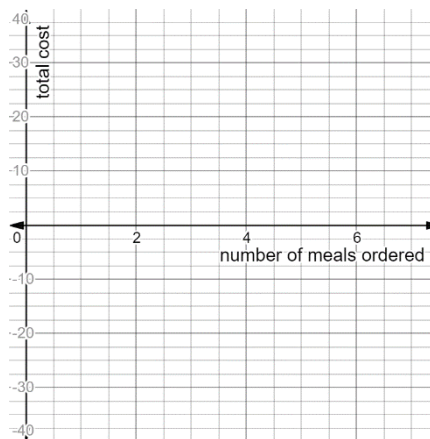
5.

Number of Meals ordered	Actual Cost	Predicted Cost	Residual Values
5	\$108		

6. How many positive residuals does the data set have?

7. What is the residual with the largest absolute value?

8. Plot the residuals on the coordinate grid.



9. Explain whether the linear function is a good fit for this data set.